Autonomous Next-Generation Aerial Refueling System (ANGARS)

Conceptual Design Report

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1.1 Physical Context Diagram

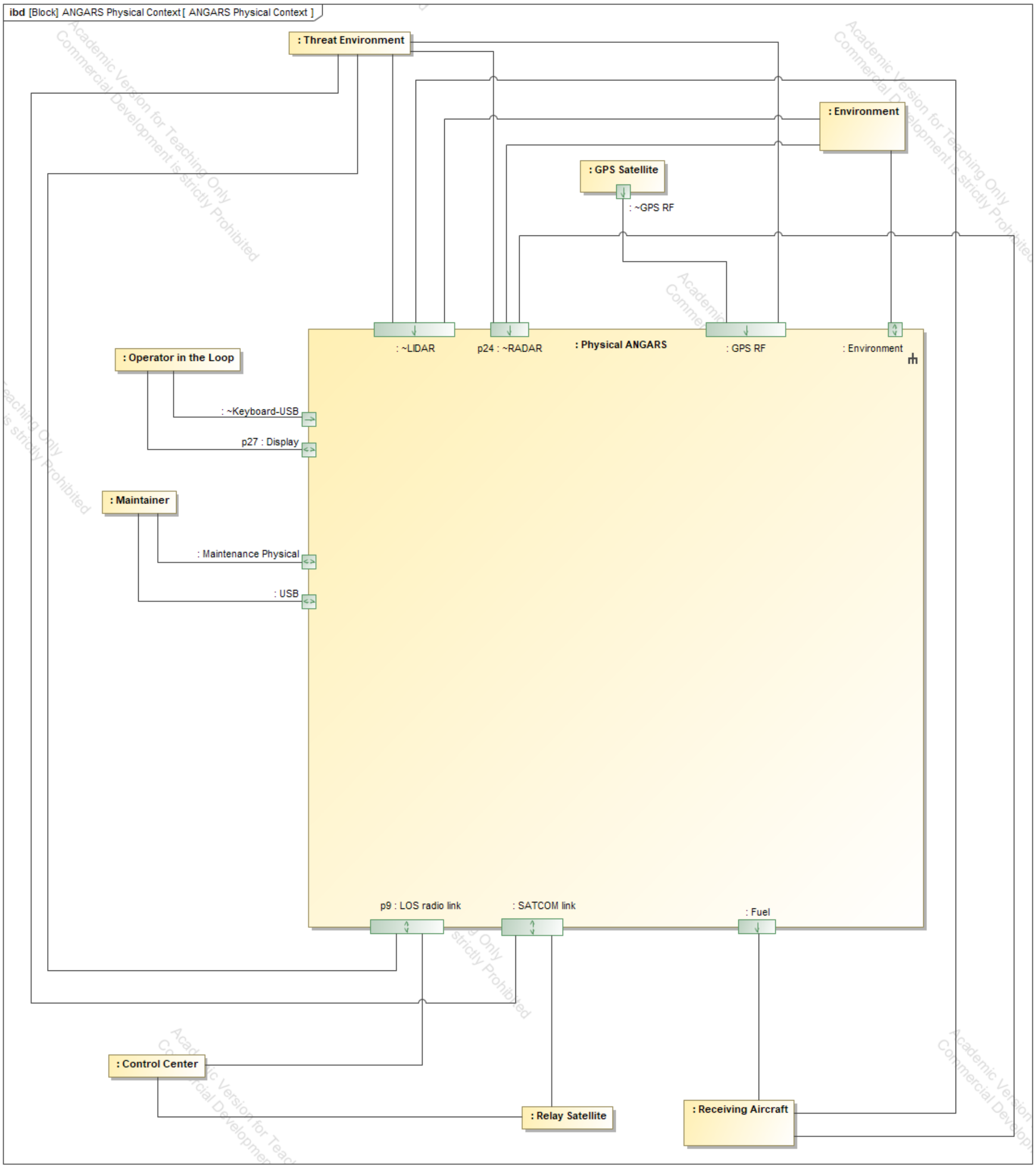

Figure 1: ANGARS Physical Context Diagram

1.2 Physical Component Tree

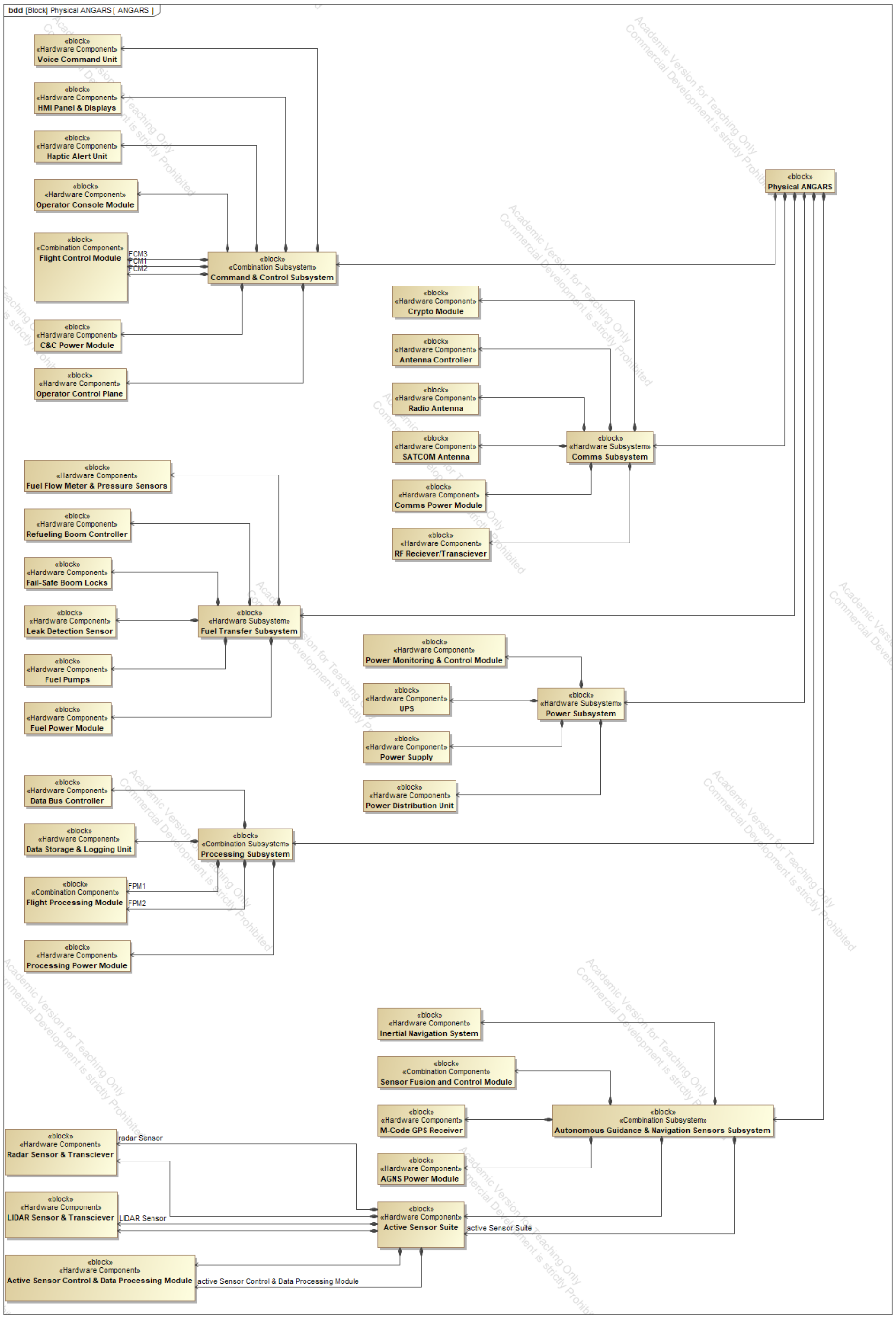

Figure 2: ANGARS Component Tree

1.3 Physical Subsystem Descriptions

Autonomous Guidance & Navigation Sensors Subsystem (AGNS)

Figure 3: AGNS Subsystem (Credit: MWRF)

This subsystem is responsible for detection and tracking of environment around the tanker, including receiving aircraft, potential adversaries, and other miscellaneous environmental factors. The subsystem enables precision approaches for refueling, alignment of receiving aircraft and tanker, and collision avoidance during refueling operations. The AGNS subsystem consists of a solution defined by the Trade Study Report. The outcome of the trade was that the AGNS subsystem solution would be that of an active suite of sensors, including; radar, LIDAR, and traditional M-code GPS. The entire sensing sensor suite is fused through a data fusion component to allow quick and efficient switching between sensors of the AGNS solution to enable the most optimal sensor available at any given time.

Command & Control (C&C) Subsystem

Figure 4: C&C Subsystem (Credit: GDMS)

This subsystem is responsible for the overall execution of mission functions. The C&C subsystem act as the main routing hub for the various different subsystems, facilitating coordination and commanding of the various peripheral subsystems. Having a central control subsystem node allows for devices which each communicate via separate protocols and formats to communicate through one common node.

The subsystem also manages safety aborts, allows for direct override from operators to get direct access to all other subsystems, providing a central location for all control needs. The subsystem contains triple-redundant flight control modules, each with independent interfaces to all C&C receiving subsystems.

Comms Subsystem

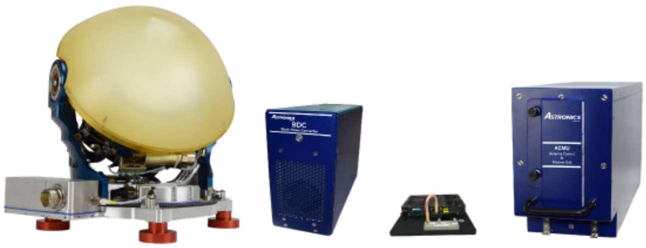

Figure 5: Comms Subsystem (Credit: astronics)

The comms subsystem is responsible for secure, low-latency, high-throughput data exchange between external actors such as receiving aircraft, ground and/or satellite relays. The subsystem employs a robust authentication approach, consisting of a crypto component before any communications can reach flight control or processing nodes, ensuring that adversaries cannot hack into the system via these paths.

The subsystem also contains a robust antenna controller capable of executing frequency hopping and other advanced anti-EW communications strategies to enable effective communication in adversarial environments. The comms subsystem has two antennas to facilitate redundant communication mechanisms.

Fuel Transfer Subsystem

The fuel transfer subsystem is responsible for control, actuation, sensing and reporting of all fuel transfer operation/boom. The fuel transfer subsystem has numerous redundancy and safety features such as multiple redundant fuel pumps, leaf detection, and automatic abort allows the system to take quick action when error events occur. Sensor data is captured and reported back through to the rest of the system to ensure system consistency, AI learning, and error diagnostic/recovery.

Figure 6: Fuel Transfer Subsystem (Credit: U.S. Air Force)

Power Subsystem

Figure 7: Power Subsystem (Credit: synqor)

The power subsystem is the heart of the ANGARS system. It regulates and disseminates power for all other subsystems. The power subsystem has redundant power supply solution including a redundant Uninterruptible Power Supply (UPS) in the event of a failure to the primary power supply. There is a common power distribution panel to allow the system to control power train enables/disables, allowing for quick subsystem power control in the event of emergency or maintenance.

The power subsystem also contains a power monitoring and control module which provides high fidelity information to the ANGARS system to provide context to the power subsystem.

Processing Subsystem

Figure 8: Processing Subsystem (Credit: militaryaerospace)

The processing subsystem is the systems brain. It is the computation backbone, hosting advanced algorithm processing, AI solutions, diagnostics, sensor fusion, data conglomeration, and more. Interfacing directly and closely with the C&C subsystem, the subsystem is the off-loaded processing node to the C&C subsystem's control. Almost all data that flows through the system is passed through the processing subsystem, making it the perfect node to store and log system data.

The processing subsystem is also responsible for mission data processing, generation, and dissemination through to the rest of the system. Redundant flight processing modules connect to the rest of the system through a common data bus, allowing either or both flight processing modules to be utilized in a scenario that requires heavy mission data processing.

1.4 Physical Block Diagrams

1.4.1 Top-Level ANGARS Block Diagram

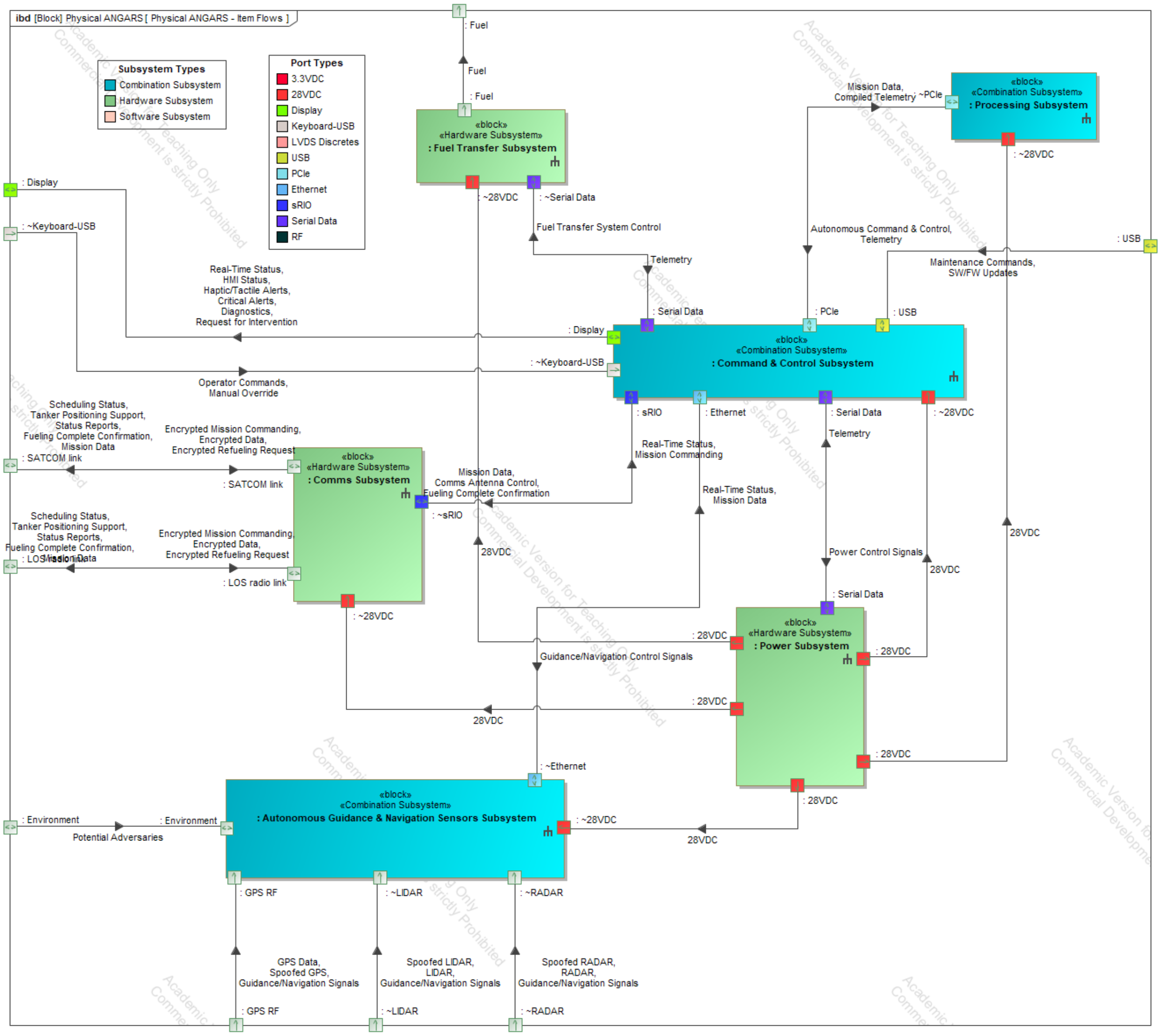

Figure 9: ANGARS Block Diagram

1.4.2 Autonomous Guidance & Navigation Sensors Subsystem Block Diagram

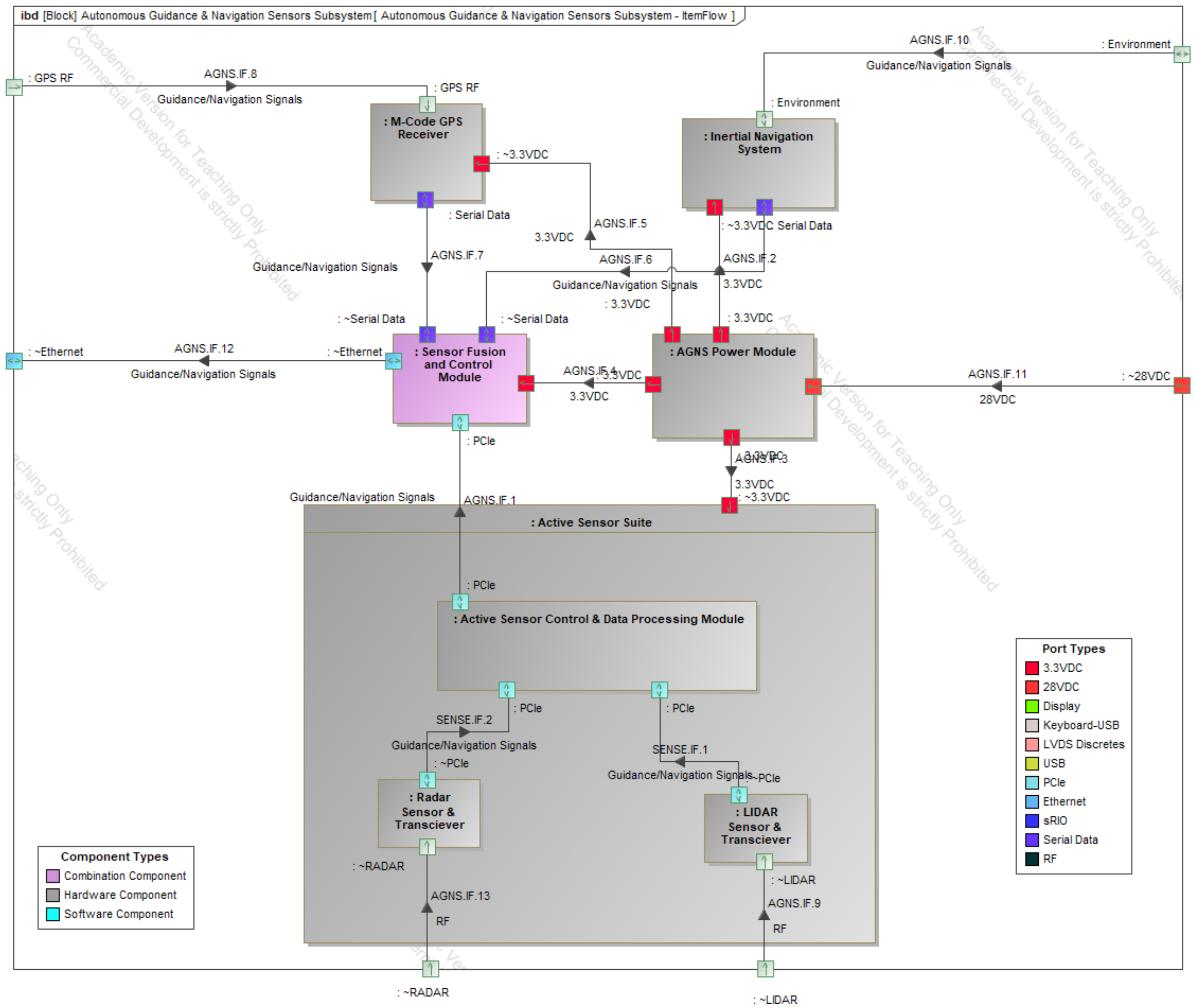

Figure 10: Autonomous Guidance & Navigation Sensors Subsystem Block Diagram

1.4.3 Command & Control Subsystem Block Diagram

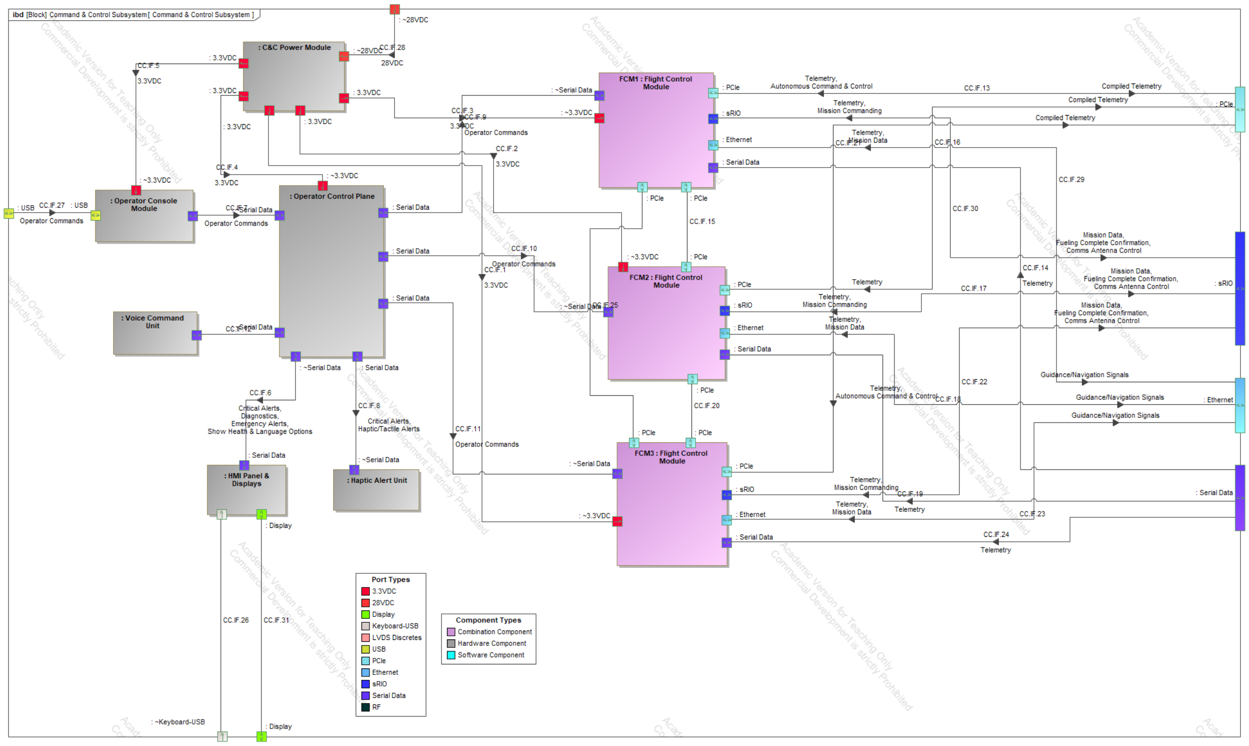

Figure 11: Command & Control Subsystem Block Diagram

1.4.4 Comms Subsystem Block Diagram

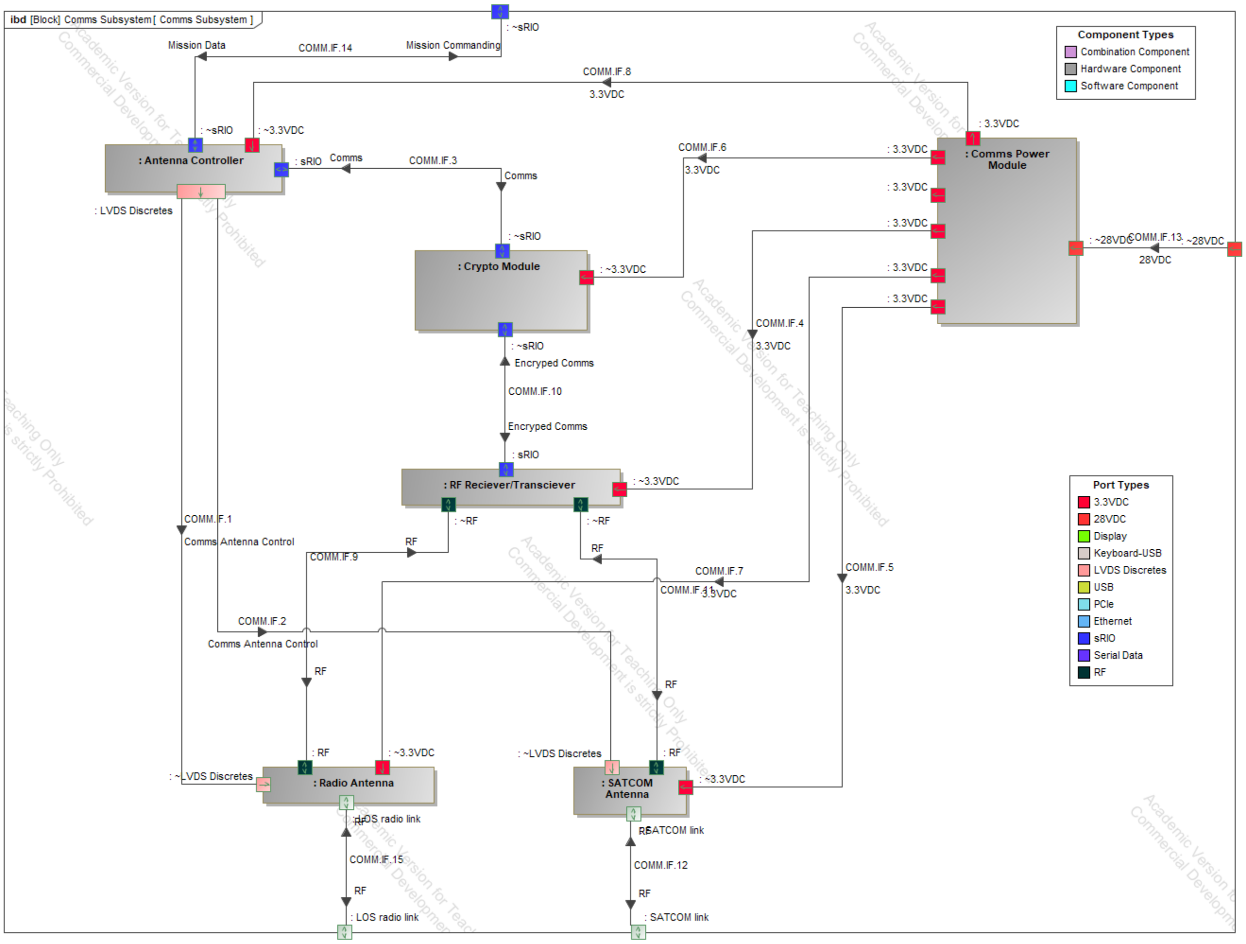

Figure 12: Comms Subsystem Block Diagram

1.4.5 Fuel Transfer Subsystem Block Diagram

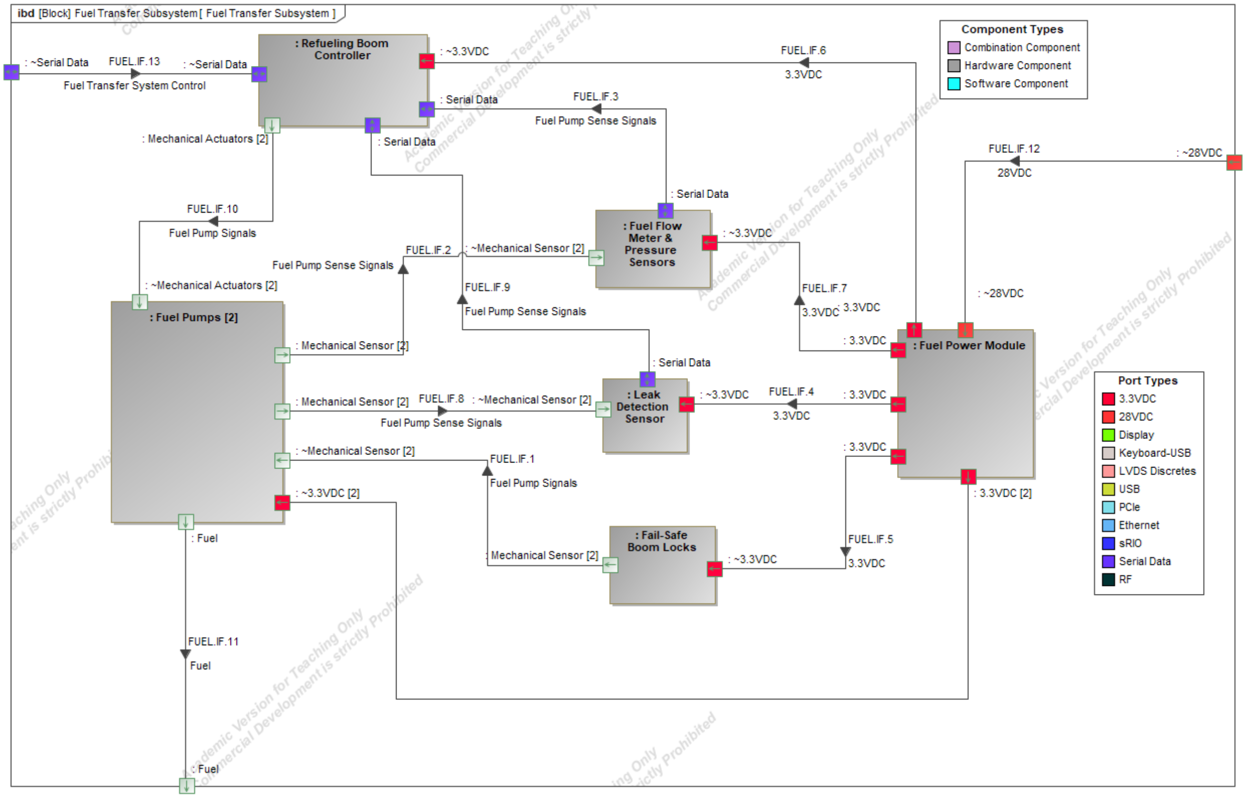

Figure 13: Fuel Transfer Subsystem Block Diagram

1.4.6 Power Subsystem Block Diagram

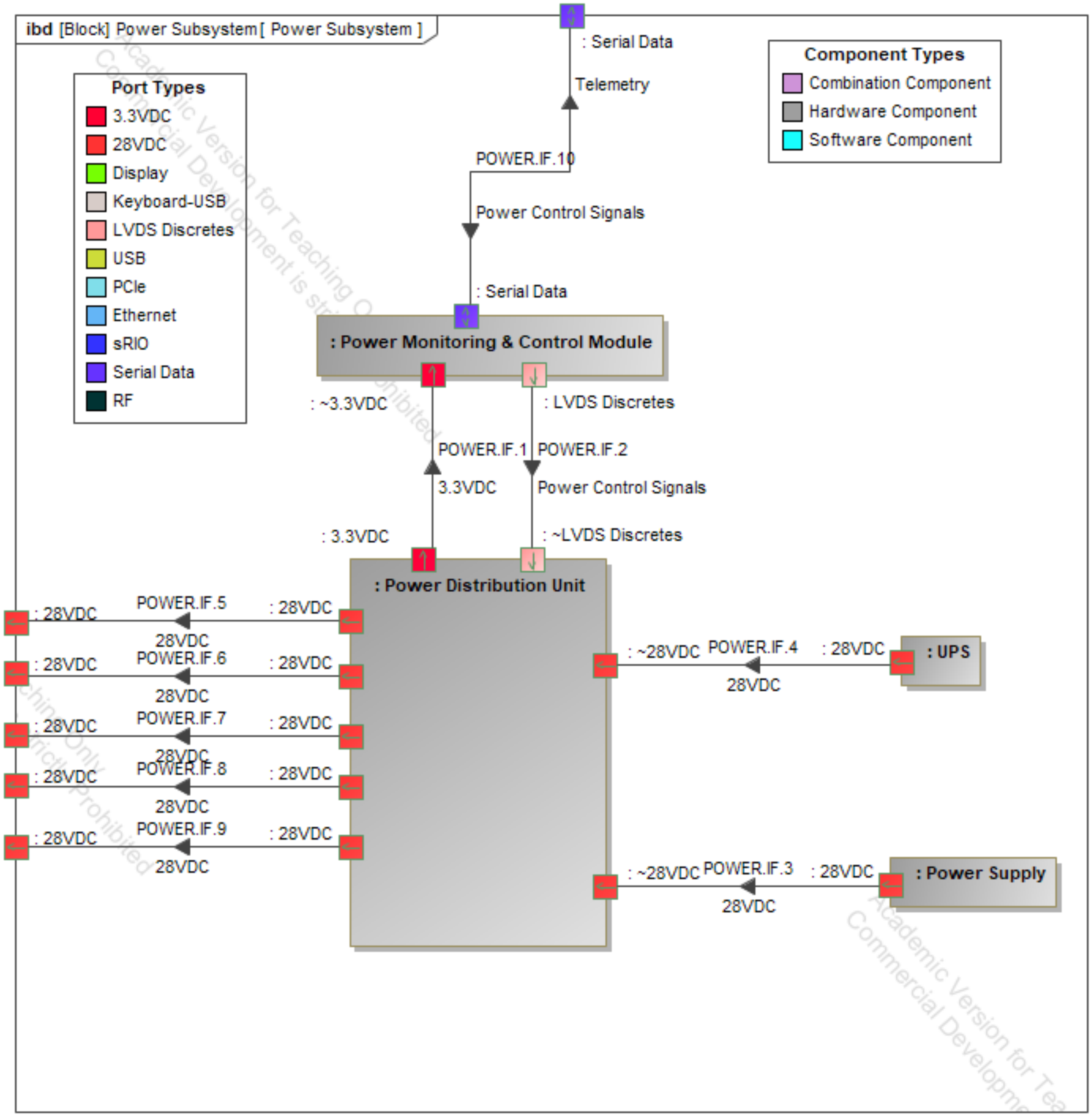

Figure 14: Power Subsystem Block Diagram

1.4.7 Processing Subsystem Block Diagram

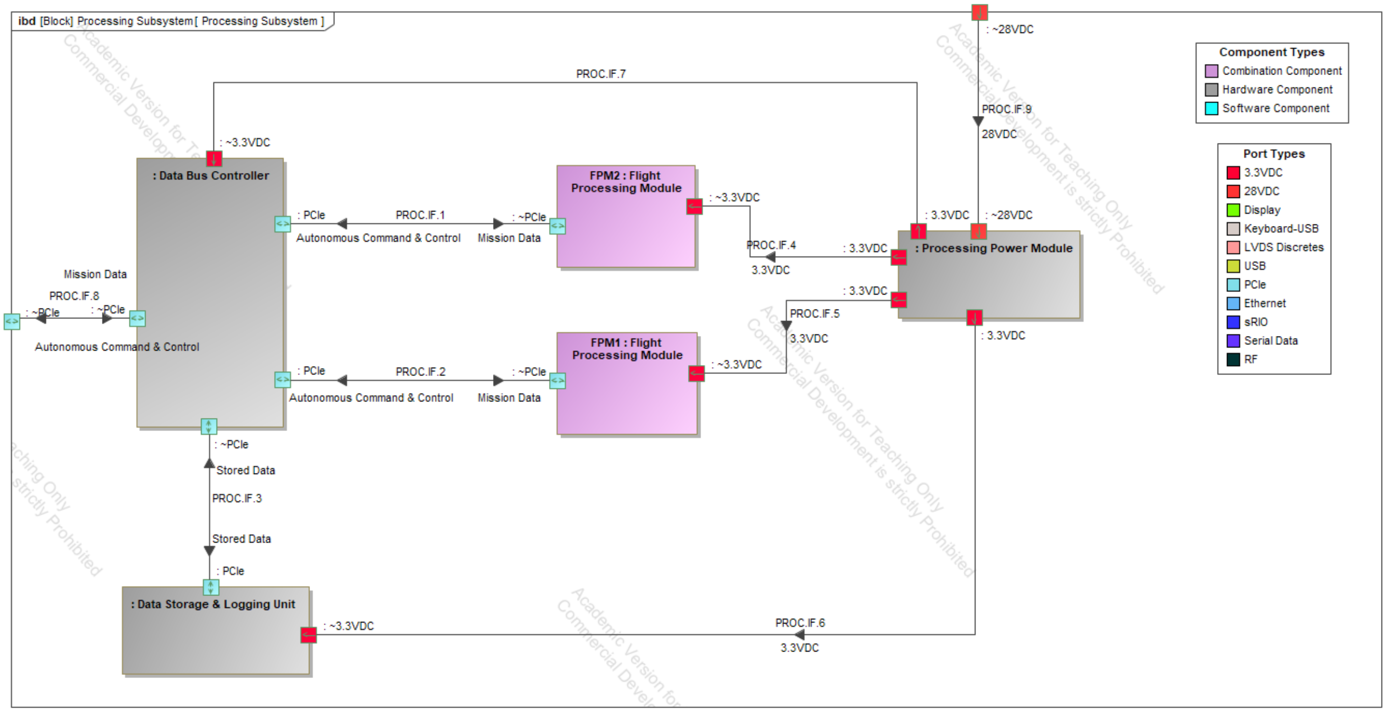

Figure 15: Processing Subsystem Block Diagram

1.5 Physical Interface Descriptions

1.5.1 Internal

Autonomous Guidance & Navigation Sensors Subsystem

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
1	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.1	Active Sensor Control & Data Processing Module	Sensor Fusion and Control Module	Guidance/Navigation Signals	PCIe	F4.1: Monitor Proximity
2	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.2	AGNS Power Module	Inertial Navigation System	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
3	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.3	AGNS Power Module	Active Sensor Suite	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
4	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.4	AGNS Power Module	Sensor Fusion and Control Module	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
5	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.5	AGNS Power Module	M-Code GPS Receiver	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
6	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.6	Inertial Navigation System	Sensor Fusion and Control Module	Guidance/Navigation Signals	Serial Data	F6.2: Analyze Environment (context Active Sensor Suite)
7	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.7	M-Code GPS Receiver	Sensor Fusion and Control Module	Guidance/Navigation Signals	Serial Data	F6.3: Analyze GPS(context Flight Processing Module)
8	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.8	Autonomous Guidance & Navigation Sensors Subsystem	M-Code GPS Receiver	Guidance/Navigation Signals	GPS RF	F6.3: Analyze GPS(context Flight Processing Module)
9	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.9	Autonomous Guidance & Navigation Sensors Subsystem	LIDAR Sensor & Transciever	RF	LIDAR	F6.2: Analyze Environment (context Active Sensor Suite)
10	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.10	Autonomous Guidance & Navigation Sensors Subsystem	Inertial Navigation System	Guidance/Navigation Signals	Environment	F6.2: Analyze Environment (context Active Sensor Suite)
11	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.11	Autonomous Guidance & Navigation Sensors Subsystem	AGNS Power Module	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
12	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.12	Sensor Fusion and Control Module	Autonomous Guidance & Navigation Sensors Subsystem	Guidance/Navigation Signals	Ethernet	F6.2: Analyze Environment (context Active Sensor Suite)
13	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.13	Autonomous Guidance & Navigation Sensors Subsystem	Radar Sensor & Transciever	RF	RADAR	F6.2: Analyze Environment (context Active Sensor Suite)

112	Active Sensor Suite	SENSE.IF.1	LIDAR Sensor & Transciever	Active Sensor Control & Data Processing Module	Guidance/Navigation Signals	PCIe	F4.1: Monitor Proximity
113	Active Sensor Suite	SENSE.IF.2	Radar Sensor & Transciever	Active Sensor Control & Data Processing Module	Guidance/Navigation Signals	PCIe	F4.1: Monitor Proximity

Command & Control Subsystem

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
							Module)
35	Command & Control Subsystem	CC.IF.2	C&C Power Module	Flight Control Module	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
36	Command & Control Subsystem	CC.IF.3	C&C Power Module	Flight Control Module	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
37	Command & Control Subsystem	CC.IF.4	C&C Power Module	Operator Control Plane	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
38	Command & Control Subsystem	CC.IF.5	C&C Power Module	Operator Console Module	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
39	Command & Control Subsystem	CC.IF.6	Operator Control Plane	HMI Panel & Displays	Critical Alerts Diagnostics Emergency Alerts Show Health & Language Options	Serial Data	F8.5: Provide Alerts & Feedback (context Haptic Alert Unit)
40	Command & Control Subsystem	CC.IF.7	Operator Console Module	Operator Control Plane	Operator Commands	Serial Data	F8.2: Receive Operator Input
41	Command & Control Subsystem	CC.IF.8	Operator Control Plane	Haptic Alert Unit	Critical Alerts Haptic/Tactile Alerts	Serial Data	F8.5: Provide Alerts & Feedback (context Haptic Alert Unit)
42	Command & Control Subsystem	CC.IF.9	Operator Control Plane	Flight Control Module	Operator Commands	Serial Data	F8.2: Receive Operator Input
43	Command & Control Subsystem	CC.IF.10	Operator Control Plane	Flight Control Module	Operator Commands	Serial Data	F8.2: Receive Operator Input
44	Command & Control Subsystem	CC.IF.11	Operator Control Plane	Flight Control Module	Operator Commands	Serial Data	F8.2: Receive Operator Input
45	Command & Control Subsystem	CC.IF.12				Serial Data	F8.8: Logging & Dashboard Integration(context HMI Panel & Displays)
46	Command & Control Subsystem	CC.IF.13	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Compiled Telemetry Autonomous Command & Control	PCIe	F5.4: Send Data
47	Command & Control Subsystem	CC.IF.14	Command & Control Subsystem	Flight Control Module	Telemetry	Serial Data	F5.4: Send Data
48	Command & Control Subsystem	CC.IF.15				PCIe	F8.8: Logging & Dashboard Integration(context HMI Panel & Displays)
49	Command & Control Subsystem	CC.IF.16	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Compiled Telemetry Autonomous Command & Control	PCIe	F5.4: Send Data

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
50	Command & Control Subsystem	CC.IF.17	Subsystem Flight Control Module	Command & Control Subsystem	Mission Commanding Mission Data Fueling Complete Confirmation Comms Antenna Control		
51	Command & Control Subsystem	CC.IF.18	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Mission Data Guidance/Navigation Signals	Ethernet	F5.4: Send Data F4.1: Monitor Proximity
52	Command & Control Subsystem	CC.IF.19	Command & Control Subsystem	Flight Control Module	Telemetry	Serial Data	F5.4: Send Data
53	Command & Control Subsystem	CC.IF.20				PCIe	F8.8: Logging & Dashboard Integration(context HMI Panel & Displays)
54	Command & Control Subsystem	CC.IF.21	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Compiled Telemetry Autonomous Command & Control	PCIe	F5.4: Send Data
55	Command & Control Subsystem	CC.IF.22	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Mission Commanding Mission Data Fueling Complete Confirmation Comms Antenna Control	sRIO	F5.4: Send Data
56	Command & Control Subsystem	CC.IF.23	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Mission Data Guidance/Navigation Signals	Ethernet	F5.4: Send Data F4.1: Monitor Proximity
57	Command & Control Subsystem	CC.IF.24	Command & Control Subsystem	Flight Control Module	Telemetry	Serial Data	F5.4: Send Data
58	Command & Control Subsystem	CC.IF.25				PCIe	F8.8: Logging & Dashboard Integration(context HMI Panel & Displays)
59	Command & Control Subsystem	CC.IF.26				Keyboard-USB	F8.2: Receive Operator Input
60	Command & Control Subsystem	CC.IF.27	Command & Control Subsystem	Operator Console Module	Operator Commands	USB	F8.2: Receive Operator Input
61	Command & Control Subsystem	CC.IF.28	Command & Control Subsystem	C&C Power Module	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
59	Command & Control Subsystem	CC.IF.26				Keyboard-USB	F8.2: Receive Operator Input
60	Command & Control Subsystem	CC.IF.27	Command & Control Subsystem	Operator Console Module	Operator Commands	USB	F8.2: Receive Operator Input
61	Command & Control Subsystem	CC.IF.28	Command & Control Subsystem	C&C Power Module	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
62	Command & Control Subsystem	CC.IF.29	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Mission Data Guidance/Navigation Signals	Ethernet	F5.4: Send Data F4.1: Monitor Proximity
63	Command & Control Subsystem	CC.IF.30	Command & Control Subsystem Flight Control Module	Flight Control Module Command & Control Subsystem	Telemetry Mission Commanding Mission Data Fueling Complete Confirmation Comms Antenna Control	sRIO	F5.4: Send Data
64	Command & Control Subsystem	CC.IF.31				Display	F8.1: Display Mission Data (context HMI Panel & Displays)

Comms Subsystem

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
66	Comms Subsystem	COMM.IF.2	Antenna Controller	SATCOM Antenna	Comms Antenna Control	LVDS Discretes	F5.4: Send Data
67	Comms Subsystem	COMM.IF.3	Antenna Controller Crypto Module	Crypto Module Antenna Controller	Comms	sRIO	F7.5: Apply Cryptography (context Crypto Module)
68	Comms Subsystem	COMM.IF.4	Comms Power Module	RF Reciever/Transciever	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
69	Comms Subsystem	COMM.IF.5	Comms Power Module	SATCOM Antenna	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
70	Comms Subsystem	COMM.IF.6	Comms Power Module	Crypto Module	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
71	Comms Subsystem	COMM.IF.7	Comms Power Module	Radio Antenna	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
72	Comms Subsystem	COMM.IF.8	Comms Power Module	Antenna Controller	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
73	Comms Subsystem	COMM.IF.9	Radio Antenna RF Reciever/Transciever	RF Reciever/Transciever Radio Antenna	RF	RF	F7.4: Zero-Trust & RF Masking (context Radio Antenna)
74	Comms Subsystem	COMM.IF.10	Crypto Module RF Reciever/Transciever	RF Reciever/Transciever Crypto Module	Encryped Comms	sRIO	F7.5: Apply Cryptography (context Crypto Module)
75	Comms Subsystem	COMM.IF.11	SATCOM Antenna RF Reciever/Transciever	RF Reciever/Transciever SATCOM Antenna	RF	RF	F7.4: Zero-Trust & RF Masking (context Radio Antenna)
76	Comms Subsystem	COMM.IF.12	Comms Subsystem SATCOM Antenna	SATCOM Antenna Comms Subsystem	RF	SATCOM link	F7.9: Send Threat Mitigation
77	Comms Subsystem	COMM.IF.13	Comms Subsystem	Comms Power Module	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
78	Comms Subsystem	COMM.IF.14	Antenna Controller Comms Subsystem	Comms Subsystem Antenna Controller	Mission Commanding Mission Data	sRIO	F5.4: Send Data
79	Comms Subsystem	COMM.IF.15	Comms Subsystem Radio Antenna	Radio Antenna Comms Subsystem	RF	LOS radio link	F7.4: Zero-Trust & RF Masking (context Radio Antenna)

Fuel Transfer Subsystem

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
80	Fuel Transfer Subsystem	FUEL.IF.1	Fail-Safe Boom Locks	Fuel Pumps	Fuel Pump Signals	Mechanical Sensor	F3.3: Initiate Fuel Transfer
81	Fuel Transfer Subsystem	FUEL.IF.2	Fuel Pumps	Fuel Flow Meter & Pressure Sensors	Fuel Pump Sense Signals	Mechanical Sensor	F3.4: Monitor Flow & Stability (context Fuel Flow Meter & Pressure Sensors)
82	Fuel Transfer Subsystem	FUEL.IF.3	Fuel Flow Meter & Pressure Sensors	Refueling Boom Controller	Fuel Pump Sense Signals	Serial Data	F3.4: Monitor Flow & Stability (context Fuel Flow Meter & Pressure Sensors)
83	Fuel Transfer Subsystem	FUEL.IF.4	Fuel Power Module	Leak Detection Sensor	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
84	Fuel Transfer Subsystem	FUEL.IF.5	Fuel Power Module	Fail-Safe Boom Locks	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
85	Fuel Transfer Subsystem	FUEL.IF.6	Fuel Power Module	Refueling Boom Controller	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
86	Fuel Transfer Subsystem	FUEL.IF.7	Fuel Power Module	Fuel Flow Meter & Pressure Sensors	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
87	Fuel Transfer Subsystem	FUEL.IF.8	Fuel Pumps	Leak Detection Sensor	Fuel Pump Sense Signals	Mechanical Sensor	F3.4: Monitor Flow & Stability (context Fuel Flow Meter & Pressure Sensors)
88	Fuel Transfer Subsystem	FUEL.IF.9	Leak Detection Sensor	Refueling Boom Controller	Fuel Pump Sense Signals	Serial Data	F3.4: Monitor Flow & Stability (context Fuel Flow Meter & Pressure Sensors)
89	Fuel Transfer Subsystem	FUEL.IF.10	Refueling Boom Controller	Fuel Pumps	Fuel Pump Signals	Mechanical Actuators	F3.7: Adjust Flow Rate
90	Fuel Transfer Subsystem	FUEL.IF.11	Fuel Pumps	Fuel Transfer Subsystem	Fuel	Fuel	F3.2: Confirm Fuel Type
91	Fuel Transfer Subsystem	FUEL.IF.12	Fuel Transfer Subsystem	Fuel Power Module	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
92	Fuel Transfer Subsystem	FUEL.IF.13	Fuel Transfer Subsystem	Refueling Boom Controller	Fuel Transfer System Control	Serial Data	F3.7: Adjust Flow Rate

Power Subsystem

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
93	Power Subsystem	POWER.IF.1	Power Distribution Unit	Power Monitoring & Control Module	3.3VDC	3.3VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
94	Power Subsystem	POWER.IF.2	Power Monitoring & Control Module	Power Distribution Unit	Power Control Signals	LVDS Discretes	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
95	Power Subsystem	POWER.IF.3	Power Supply	Power Distribution Unit	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
96	Power Subsystem	POWER.IF.4	UPS	Power Distribution Unit	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
97	Power Subsystem	POWER.IF.5	Power Distribution Unit	Power Subsystem	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
98	Power Subsystem	POWER.IF.6	Power Distribution Unit	Power Subsystem	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
99	Power Subsystem	POWER.IF.7	Power Distribution Unit	Power Subsystem	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
100	Power Subsystem	POWER.IF.8	Power Distribution Unit	Power Subsystem	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
101	Power Subsystem	POWER.IF.9	Power Distribution Unit	Power Subsystem	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
102	Power Subsystem	POWER.IF.10	Power Subsystem Power Monitoring & Control Module	Power Monitoring & Control Module Power Subsystem	Power Control Signals Telemetry	Serial Data	**F5.4: Send Data**

Processing Subsystem

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
103	Processing Subsystem	PROC.IF.1	Data Bus Controller Flight Processing Module	Flight Processing Module Data Bus Controller	Mission Data Autonomous Command & Control	PCIe	**F5.4: Send Data**
104	Processing Subsystem	PROC.IF.2	Data Bus Controller Flight Processing Module	Flight Processing Module Data Bus Controller	Mission Data Autonomous Command & Control	PCIe	**F5.4: Send Data**
105	Processing Subsystem	PROC.IF.3	Data Bus Controller Data Storage & Logging Unit	Data Storage & Logging Unit Data Bus Controller	Stored Data	PCIe	**F8.8: Logging & Dashboard Integration**(context HMI Panel & Displays)
106	Processing Subsystem	PROC.IF.4	Processing Power Module	Flight Processing Module	3.3VDC	3.3VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
107	Processing Subsystem	PROC.IF.5	Processing Power Module	Flight Processing Module	3.3VDC	3.3VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
108	Processing Subsystem	PROC.IF.6	Processing Power Module	Data Storage & Logging Unit	3.3VDC	3.3VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
109	Processing Subsystem	PROC.IF.7				3.3VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)

1.5.2 External

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
1	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.1	Active Sensor Control & Data Processing Module	Sensor Fusion and Control Module	Guidance/Navigation Signals	PCIe	F4.1: Monitor Proximity
2	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.2	AGNS Power Module	Inertial Navigation System	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
3	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.3	AGNS Power Module	Active Sensor Suite	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
4	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.4	AGNS Power Module	Sensor Fusion and Control Module	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
5	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.5	AGNS Power Module	M-Code GPS Receiver	3.3VDC	3.3VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
6	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.6	Inertial Navigation System	Sensor Fusion and Control Module	Guidance/Navigation Signals	Serial Data	F6.2: Analyze Environment (context Active Sensor Suite)
7	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.7	M-Code GPS Receiver	Sensor Fusion and Control Module	Guidance/Navigation Signals	Serial Data	F6.3: Analyze GPS(context Flight Processing Module)
8	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.8	Autonomous Guidance & Navigation Sensors Subsystem	M-Code GPS Receiver	Guidance/Navigation Signals	GPS RF	F6.3: Analyze GPS(context Flight Processing Module)
9	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.9	Autonomous Guidance & Navigation Sensors Subsystem	LIDAR Sensor & Transciever	RF	LIDAR	F6.2: Analyze Environment (context Active Sensor Suite)
10	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.10	Autonomous Guidance & Navigation Sensors Subsystem	Inertial Navigation System	Guidance/Navigation Signals	Environment	F6.2: Analyze Environment (context Active Sensor Suite)
11	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.11	Autonomous Guidance & Navigation Sensors Subsystem	AGNS Power Module	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
12	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.12	Sensor Fusion and Control Module	Autonomous Guidance & Navigation Sensors Subsystem	Guidance/Navigation Signals	Ethernet	F6.2: Analyze Environment (context Active Sensor Suite)
13	Autonomous Guidance & Navigation Sensors Subsystem	AGNS.IF.13	Autonomous Guidance & Navigation Sensors Subsystem	Radar Sensor & Transciever	RF	RADAR	F6.2: Analyze Environment (context Active Sensor Suite)
14	Physical ANGARS	ANGARS.IF.1	Physical ANGARS	Autonomous Guidance & Navigation Sensors Subsystem	Spoofed LIDAR LIDAR Guidance/Navigation Signals	LIDAR	F6.1: Alert System(context Haptic Alert Unit)
15	Physical ANGARS	ANGARS.IF.2	Autonomous Guidance & Navigation Sensors Subsystem Command & Control Subsystem	Command & Control Subsystem Autonomous Guidance & Navigation Sensors Subsystem	Real-Time Status Mission Data Guidance/Navigation Control Signals	Ethernet	F5.2: Receive Mission Data F4.1: Monitor Proximity

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
					Signals		
15	Physical ANGARS	ANGARS.IF.2	Autonomous Guidance & Navigation Sensors Subsystem Command & Control Subsystem	Command & Control Subsystem Autonomous Guidance & Navigation Sensors Subsystem	Real-Time Status Mission Data Guidance/Navigation Control Signals	Ethernet	F5.2: Receive Mission Data F4.1: Monitor Proximity
16	Physical ANGARS	ANGARS.IF.3	Physical ANGARS	Autonomous Guidance & Navigation Sensors Subsystem	Spoofed RADAR RADAR Guidance/Navigation Signals	RADAR	F6.1: Alert System(context Haptic Alert Unit)
17	Physical ANGARS	ANGARS.IF.4	Physical ANGARS	Command & Control Subsystem	Maintenance Commands SW/FW Updates	USB	F5.3: Receive Maintenance Operations and Files
18	Physical ANGARS	ANGARS.IF.5	Command & Control Subsystem Fuel Transfer Subsystem	Fuel Transfer Subsystem Command & Control Subsystem	Fuel Transfer System Control Telemetry	Serial Data	F3.3: Initiate Fuel Transfer
19	Physical ANGARS	ANGARS.IF.6	Command & Control Subsystem	Physical ANGARS	Real-Time Status HMI Status Haptic/Tactile Alerts Critical Alerts Diagnostics Request for Intervention	Display	F8.5: Provide Alerts & Feedback (context Haptic Alert Unit)
20	Physical ANGARS	ANGARS.IF.7	Power Subsystem	Comms Subsystem	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
21	Physical ANGARS	ANGARS.IF.8	Comms Subsystem Command & Control Subsystem	Command & Control Subsystem Comms Subsystem	Real-Time Status Mission Data Comms Antenna Control Fueling Complete Confirmation Mission Commanding	sRIO	F5.2: Receive Mission Data
22	Physical ANGARS	ANGARS.IF.9	Fuel Transfer Subsystem	Physical ANGARS	Fuel	Fuel	F3.2: Confirm Fuel Type
23	Physical ANGARS	ANGARS.IF.10	Power Subsystem	Processing Subsystem	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)
24	Physical ANGARS	ANGARS.IF.11	Power Subsystem	Command & Control Subsystem	28VDC	28VDC	F9.1 Distribute Power(context Power Monitoring & Control Module)

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
							Module)
25	Physical ANGARS	ANGARS.IF.12	Power Subsystem	Fuel Transfer Subsystem	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
26	Physical ANGARS	ANGARS.IF.13	Power Subsystem	Autonomous Guidance & Navigation Sensors Subsystem	28VDC	28VDC	**F9.1 Distribute Power**(context Power Monitoring & Control Module)
27	Physical ANGARS	ANGARS.IF.14	Power Subsystem Command & Control Subsystem	Command & Control Subsystem Power Subsystem	Telemetry Power Control Signals	Serial Data	**F5.4: Send Data**
28	Physical ANGARS	ANGARS.IF.15	Command & Control Subsystem Processing Subsystem	Processing Subsystem Command & Control Subsystem	Mission Data Autonomous Command & Control Telemetry Compiled Telemetry	PCIe	**F5.4: Send Data**
29	Physical ANGARS	ANGARS.IF.16	Physical ANGARS	Command & Control Subsystem	Operator Commands Manual Override	Keyboard-USB	**F8.2: Receive Operator Input**
30	Physical ANGARS	ANGARS.IF.17	Physical ANGARS	Autonomous Guidance & Navigation Sensors Subsystem	Potential Adversaries	Environment	**F6.1: Alert System**(context Haptic Alert Unit)
31	Physical ANGARS	ANGARS.IF.18	Physical ANGARS Comms Subsystem	Comms Subsystem Physical ANGARS	Encrypted Mission Commanding Scheduling Status Tanker Positioning Support Status Reports Fueling Complete Confirmation Mission Data Encrypted Data Encrypted Refueling Request	SATCOM link	**F7.5: Apply Cryptography** (context Crypto Module)
32	Physical ANGARS	ANGARS.IF.19	Physical ANGARS	Autonomous Guidance & Navigation Sensors Subsystem	GPS Data Spoofed GPS Guidance/Navigation Signals	GPS RF	**F6.3: Analyze GPS**(context Flight Processing Module)

Physical ANGARS [3.0 Physical Architecture::3.2 Physical Elements]

#	Owner	Name	Source	Target	Conveyed	Type	Allocated To
33	Physical ANGARS	ANGARS.IF.20			Scheduling Status Tanker Positioning Support Status Reports Fueling Complete Confirmation Mission Data Encrypted Data Encrypted Refueling Request		

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1.6 Physical N2

1.6.1 Subsystem

Source / Destination	External	Autonomous Guidance & Navigation Sensors Subsystem	Command & Control Subsystem	Fuel Transfer Subsystem	Power Subsystem	Comms Subsystem	Processing Subsystem
External		Spoofed LIDAR, LIDAR, Guidance/Navigation Signals, Spoofed RADAR, RADAR, Guidance/Navigation Signals, Potential Adversaries, GPS Data, Spoofed GPS, Guidance/Navigation Signals	Maintenance Commands, SW/FW Updates, Operator Commands, Manual Override			Encrypted Mission Commanding, Scheduling Status, Tanker Positioning Support, Status Reports, Fueling Complete Confirmation, Mission Data, Encrypted Data, Encrypted Refueling Request	
Autonomous Guidance & Navigation Sensors Subsystem			Real-Time Status, Mission Data, Guidance/Navigation Control Signals				
Command & Control Subsystem	Real-Time Status, HMI Status, Haptic/Tactile Alerts, Critical Alerts, Diagnostics, Request for Intervention	Real-Time Status, Mission Data, Guidance/Navigation Control Signals		Fuel Transfer System Control, Telemetry			Mission Data, Autonomous Command & Control, Telemetry, Compiled Telemetry
Fuel Transfer Subsystem	Fuel		Fuel Transfer System Control, Telemetry				
Power Subsystem		28VDC	28VDC, Telemetry, Power Control Signals	28VDC		28VDC	28VDC
Comms Subsystem			Real-Time Status, Mission Data, Comms Antenna Control, Fueling Complete Confirmation, Mission Commanding				
Processing Subsystem			Mission Data, Autonomous Command & Control, Telemetry, Compiled Telemetry				

Figure 16: Physical Subsystem N2

1.6.2 Components

AGNS Subsystem

Source / Destination	External	LIDAR Sensor & Transciever	Radar Sensor & Transciever	Active Sensor Control & Data Processing Module	Sensor Fusion and Control Module	AGNS Power Module	Inertial Navigation System	Active Sensor Suite	MCode GPS Receiver
External		RF	RF			28VDC	Guidance/Navigation Signals		Guidance/Navigation Signals
LIDAR Sensor & Transciever				Guidance/Navigation Signals					
Radar Sensor & Transciever				Guidance/Navigation Signals					
Active Sensor Control & Data Processing Module					Guidance/Navigation Signals				
Sensor Fusion and Control Module	Guidance/Navigation Signals								
AGNS Power Module					3.3VDC		3.3VDC	3.3VDC	3.3VDC
Inertial Navigation System					Guidance/Navigation Signals				
Active Sensor Suite									
MCode GPS Receiver					Guidance/Navigation Signals				

Figure 17: AGNS Subsystem N2

Command & Control Subsystem

Source / Destination	External	C&C Power Module	Operator Control Plane	Operator Console Module	HMI Panel & Displays	Haptic Alert Unit	Flight Control Module
External		28VDC		Operator Commands			Telemetry, Mission Commanding, Mission Data, Fueling Complete Confirmation, Comms Antenna Control, Guidance/Navigation Signals
C&C Power Module			3.3VDC	3.3VDC			3.3VDC
Operator Control Plane					Critical Alerts, Diagnostics, Emergency Alerts, Show Health & Language Options	Critical Alerts, Haptic/Tactile Alerts	Operator Commands
Operator Console Module			Operator Commands				
HMI Panel & Displays							
Haptic Alert Unit							
Flight Control Module	Telemetry, Compiled Telemetry, Autonomous Command & Control, Mission Data, Guidance/Navigation Signals						

Figure 18: Command & Control Subsystem N2

Comms Subsystem

Source / Destination	External	Antenna Controller	Radio Antenna	SATCOM Antenna	Crypto Module	RF Reciever/Transciever	Comms Power Module
External		Mission Commanding, Mission Data	RF	RF			28VDC
Antenna Controller	Mission Commanding, Mission Data		Comms Antenna Control	Comms Antenna Control	Comms		
Radio Antenna	RF					RF	
SATCOM Antenna	RF					RF	
Crypto Module		Comms				Encrypted Comms	
RF Reciever/Transciever			RF	RF	Encrypted Comms		
Comms Power Module		3.3VDC	3.3VDC	3.3VDC	3.3VDC	3.3VDC	

Figure 19: Comms Subsystem N2

Fuel Transfer Subsystem

Source / Destination	External	Fail-Safe Boom Locks	Fuel Pump	Fuel Flow Meter & Pressure Sensors	Fuel Power Module	Leak Detection Sensor	Refueling Boom Controller
External					28VDC		Fuel Transfer System Control
Fail-Safe Boom Locks			Fuel Pump Signals				
Fuel Pump	Fuel			Fuel Pump Sense Signals		Fuel Pump Sense Signals	
Fuel Flow Meter & Pressure Sensors							Fuel Pump Sense Signals
Fuel Power Module		3.3VDC		3.3VDC		3.3VDC	3.3VDC
Leak Detection Sensor							Fuel Pump Sense Signals
Refueling Boom Controller			Fuel Pump Signals				

Figure 20: Fuel Transfer Subsystem N2

Power Subsystem

Source / Destination	External	Power Supply	UPS	Power Distribution Unit	Power Monitoring & Control Module
External					Power Control Signals, Telemetry
Power Supply				28VDC	
UPS				28VDC	
Power Distribution Unit	28VDC				3.3VDC
Power Monitoring & Control Module	Power Control Signals, Telemetry			Power Control Signals	

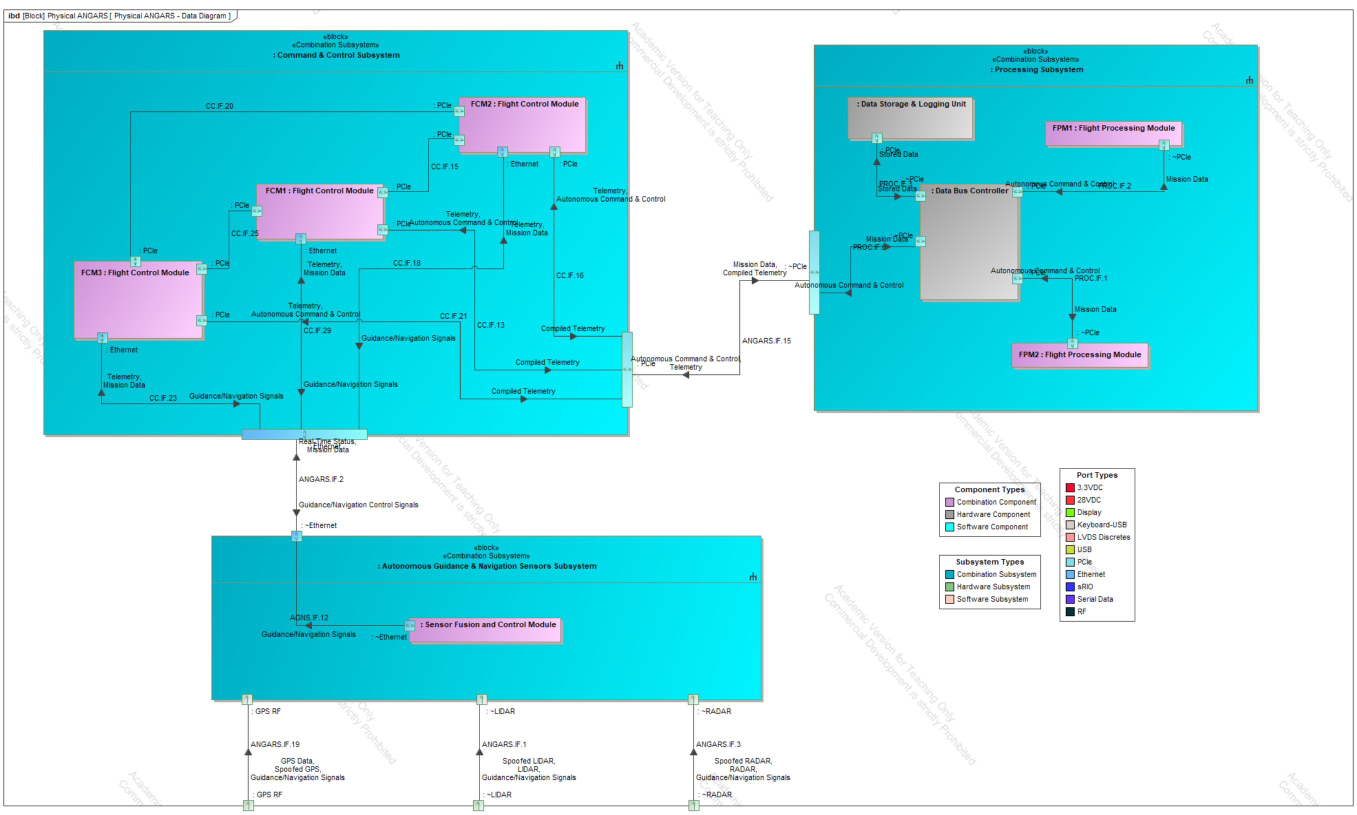
Figure 21: Power Subsystem N2

Processing Subsystem

Source / Destination	External	Data Bus Controller	Flight Processing Module	Data Storage & Logging Unit	Processing Power Module
External		Mission Data, Autonomous Command & Control			28VDC
Data Bus Controller	Mission Data, Autonomous Command & Control		Mission Data, Autonomous Command & Control	Stored Data	
Flight Processing Module		Mission Data, Autonomous Command & Control			
Data Storage & Logging Unit		Stored Data			
Processing Power Module			3.3VDC	3.3VDC	

Figure 22: Processing Subsystem N2

1.7 Data Flow Diagram



1.8 Data Dictionary

#	Name	Documentation	Format
1	3.3VDC	Regulated 3.3V power supply for powering lower voltage components.	VDC
2	28VDC	Regulated 28V power supply for power sources.	VDC
3	Autonomous Command & Control	Autonomous command and control instructions. Enables the system to perform operations independently without manual input.	PCIe
4	Comms	System-wide communication data, typically sourced from or to the ground. Facilitates the exchange of information between various subsystems.	sRIO
5	Comms Antenna Control	Governs orientation and switching of antennas.	LVDS Discretes
6	Compiled Telemetry	Aggregates multiple telemetry data streams into one feed. Simplifies monitoring by providing a consolidated lens of sensor outputs.	PCIe
7	Critical Alerts	Indicator of high-priority alerts that require immediate attention. Notifies operators of urgent system issues.	Display
8	Diagnostics	Provides diagnostic data for monitoring system health. Captures and logs performance metrics and operational status information.	Display
9	Emergency Alerts	Indicator of emergency conditions during critical events. Notifies operators when immediate response is required.	Display
10	Encryped Comms	Provides secure encrypted communications.	sRIO
11	Encrypted Data	Encrypted mission data transmission. Safeguards sensitive information during operations.	SATCOM link
12	Encrypted Mission Commanding	Secure mission command instructions. Ensures commands remain confidential during transmission.	SATCOM link
13	Encrypted Refueling Request	Encrypted request for refueling operations. Validates the authenticity of refueling commands.	SATCOM link
14	Fuel	Actual fuel flowing out of system.	Fuel
15	Fuel Pump Sense Signals	Sensor feedback data from fuel pumps. Monitors the operational status and performance of fuel pump sensors.	Mechanical Sensor
16	Fuel Pump Signals	Activation and control signals for the fuel pumps.	Mechanical Actuators
17	Fuel Transfer System Control	Control signal for regulating fuel transfer operations. Coordinates the overall fuel transfer process.	Serial Data
18	Fueling Complete Confirmation	Confirmation signal indicating completion of fuel transfer. Verifies that fueling operations have concluded successfully.	sRIO
19	GPS Data	GPS positioning and timing information. Provides data essential for navigation and location information.	GPS RF
20	Guidance/Navigation Control Signals	Guidance control command data. Directs system movements along the path.	Ethernet
21	Guidance/Navigation Signals	Navigation sensor data for positional awareness. Provides inputs for determining system orientation.	PCIe
22	Haptic/Tactile Alerts	Tactile feedback alert signal. Notifies users through physical cues during specific conditions.	Serial Data
23	HMI Status	Status data from the Human-Machine Interface. Displays the current operational state of the HMI.	Display
24	LIDAR	LIDAR mapping data signal. Used for understanding spatial information for object detection and navigation.	LIDAR
25	Maintenance Commands	Signal carrying maintenance instructions. Initiates and coordinates maintenance operations across the system.	USB
26	Manual Override	Manual control override signal. Allows operators to bypass automated processes when necessary.	Keyboard-USB
27	Mission Commanding	Mission command instructions signal. Directs the system to execute mission-specific tasks.	sRIO
28	Mission Data	Transmission of mission-specific data. Contains critical information required for mission success.	Ethernet
29	Operator Commands	Operator-issued command signal. Allows manual adjustments and controls during active operations.	Serial Data
30	Potential Adversaries	Detection signal for potential adversarial threats. Alerts the system of possible risks from external sources.	Environment
31	Power Control Signals	Signals managing power distribution across the system. Coordinates allocation and control of electrical power to various components.	LVDS Discretes
32	RADAR	Radar data signal. Captures raw radar returns for object identification and tracking.	RADAR
33	Real-Time Status	Real-time operational status data. Continuously updates system status and health information.	Display
34	Request for Intervention	Signal prompting operator intervention	Display
35	RF	Radio frequency comms data. Enables wireless data transmission and reception.	RF
36	Scheduling Status	Scheduling and timing data signal. Ensures proper coordination and synchronization of system operations.	SATCOM link
37	Show Health & Language Options	Display signal for system health and language options. Provides users with current operational status and configuration settings.	Display
38	Spoofed GPS	Adversarial GPS data signal. Signal generated from adversarial source in hopes of deception.	GPS RF
39	Spoofed LIDAR	Adversarial LIDAR data signal. Signal generated from adversarial source in hopes of deception.	LIDAR
40	Spoofed RADAR	Adversarial RADAR data signal. Signal generated from adversarial source in hopes of deception.	RADAR
41	Status Reports	Compilation of system status reports. Aggregates status and health data for analysis.	sRIO
42	Stored Data	Archived data signals. Stores historical information for future review and analysis.	PCIe
43	SW/FW Updates	Signal containing software and firmware update data.	USB
44	Tanker Positioning Support	Support signal for tanker positioning operations. Provides essential guidance for accurate tanker alignment.	SATCOM link
45	Telemetry	Telemetry data transmission signal. Conveys detailed sensor measurements and operational information for monitoring purposes.	PCIe
46	Encrypted Comms	Secure, encrypted communication data. Ensures confidentiality of transmitted messages.	Ethernet

1.9 Behavioral to Structural Allocation

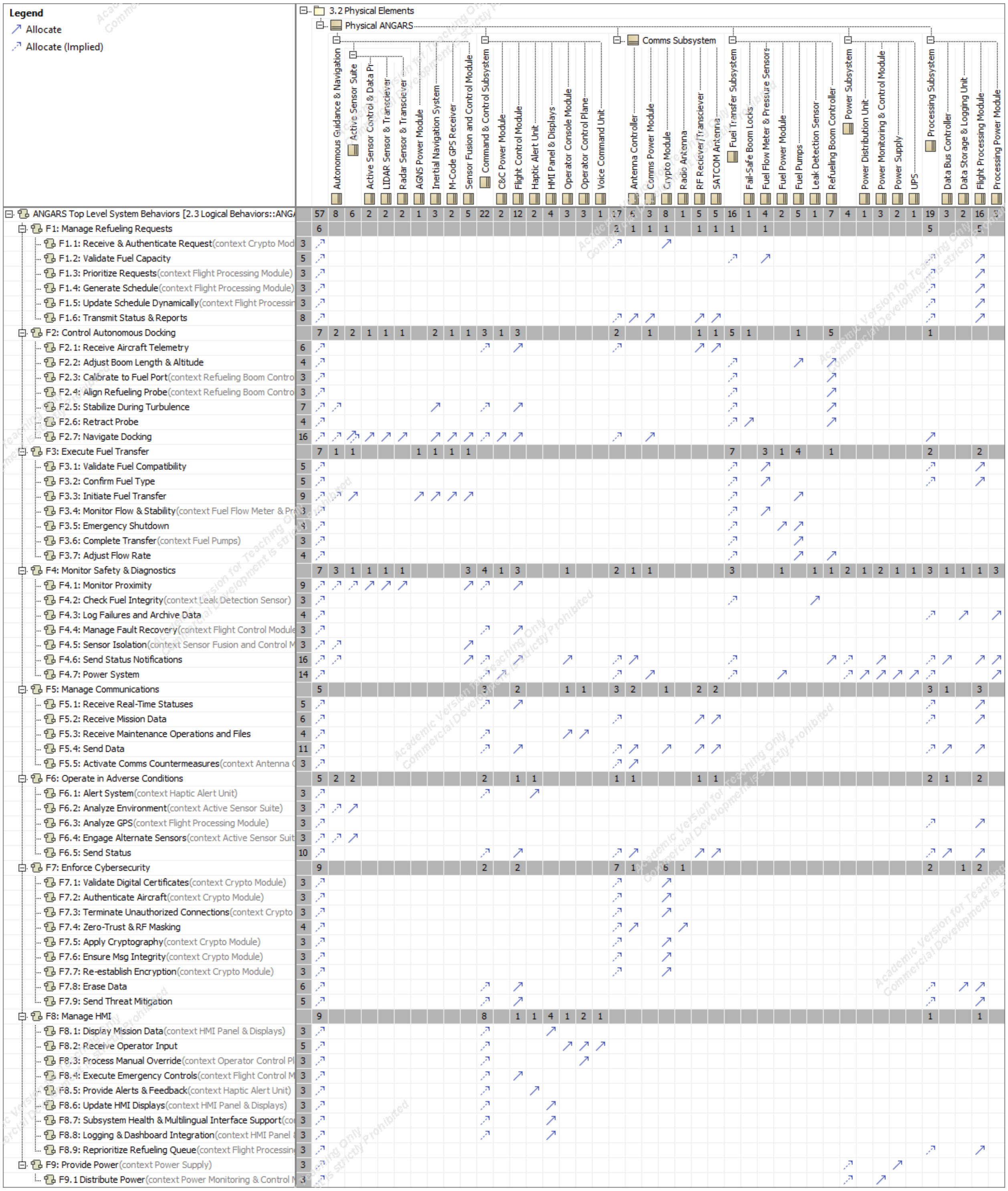


Figure 23: Structural to Behavioral Allocation Matrix

2 Specification Updates

2.1 ANGARS Requirements Specification Updates

As systems engineering is an iterative process, requirements are continually being refined and updated to ultimately

Table 1: Requirements Tracker

	Total	Quantitative	%	Binary	Qualitative
Requirements Analysis Report	154	48	31.1%	63	43
Functional Analysis Report	154	52	33.7%	63	39
Trade Study	159	56	35.2%	63	39
Conceptual Design Report	165	62	37.6%	63	40
Test Plan					
System Specifications					

2.2 Added Requirements

#	Id	Name	Text	Derived From	classification	Verify Method	KPP
1	ANGARS-157	ANGARS-157 Subsystem Backup Activation Time	The system shall activate backup subsystems within 2 seconds (T), 1 second (O) of primary subsystem failure.	N6 Redundant Fail-Safe Systems N7 Mission Reliability	Quantitative	Analysis	false
2	ANGARS-158	ANGARS-158 Backup sensing and navigation	The system shall have one (T), three (O) distinct navigation system redundancy modes.	N6 Redundant Fail-Safe Systems N7 Mission Reliability	Quantitative	Analysis	false
3	ANGARS-159	ANGARS-159 Antijam resistance	The system shall maintain functionality through 55 dBm (T), 65 dBm (O) jamming attempts.	N3 Contested Environment Operation	Quantitative	Test	false
4	ANGARS-160	ANGARS-160 Sensor Fusion Switch Time	The system shall automatically switch between active sensor inputs (radar, LIDAR, and M-code GPS) within 1.5 seconds (T), 1.0 second (O) upon detecting sensor performance degradation.	N3 Contested Environment Operation	Quantitative	Test	false
5	ANGARS-161	ANGARS-161 Flight Control Synchronization	The system shall ensure that the triple-redundant flight control modules synchronize command execution within 100 ms (T), 50 ms (O) of each other.	N3 Contested Environment Operation N6 Redundant Fail-Safe Systems	Quantitative	Test	false
6	ANGARS-162	ANGARS-162 Antenna Redundancy Failover Time	The system shall detect a primary antenna degredation and switch to the secondary antenna within 500 ms (T), 300 ms (O) of failure detection.	N3 Contested Environment Operation N4 Secure Communication	Quantitative	Test	false
7	ANGARS-163	ANGARS-163 Fuel Pump Redundancy Activation Time	The system shall activate a redundant fuel pump and maintain continuous fuel transfer within 3 seconds (T), 2 seconds (O) of detecting a primary pump fault.	N8 Health Monitoring & Diagnostics	Quantitative	Test	false
8	ANGARS-164	ANGARS-164 Power Subsystem Failover Time	The system shall switch to backup UPS and distribution within 2 seconds (T), 1 second (O) in the event of a primary power failure.	N6 Redundant Fail-Safe Systems N7 Mission Reliability N10 Threat Detection & Response	Quantitative	Test	false
9	ANGARS-165	ANGARS-165 Processing Data Fusion Latency	The system shall fuse and process sensor data from the AGNS subsystem and update system status within 500 ms (T), 300 ms (O) under nominal operational conditions.	N6 Redundant Fail-Safe Systems N7 Mission Reliability N10 Threat Detection & Response	Quantitative	Test	false

Appendix A – Schedule and EVM Updates

Schedule

Table 2: Capstone Schedule

WBS ID	Task Name	Planned Work Hours	Planned Milestone Start	Planned Milestone Finish	Actual Work Hours	Actual Milestone Start	Actual Milestone Finish
1	**Project Concept & Proposal**	50	10-Dec	21-Jan	25	10-Dec	1/19/2025
1.1	Intro	5					
1.3	System Need	5					
1.4	System Description	15					
1.5	Background	10					
1.6	Justification	5					
1.7	Draft Cleanup & Submittal	5					
1.8	Final Revisions & Submittal	5					
2	**Requirements Analysis Report (RAR) & CONOPS**	45	21-Jan	2-Feb		21-Jan	2-Feb
2.1	Requirements Definition & Description	8					
2.2	Requirements Verification Plan	6					
2.3	User/Stakeholder Needs Anaylsis & Report	7					
2.4	Key Performance Parameters (KPP) Identification & Compilation	6					
2.5	CONOPS - Concept Diagram	6					
2.6	CONOPS - Potential Users/Operators List	4					
2.7	CONOPS - Use Case Scenario Definition	5					
2.8	RAR & CONOPS Cleanup & Submittal	3					
3	**Functional Analysis Report (FAR)**	37	2-Feb	16-Feb			
3.1	Functional Tree Definition	5					
3.2	Functional Decomp / Tree	5					
3.3	Context Diagram UPDATES	3					
3.4	Function List Development	4					
3.5	Functional Flow Diagrams Creation	6					
3.6	N2 Diagrams Development	5					
3.7	Requirements-Functions Traceability Matrix	4					
3.8	System States & Modes Definition	3					
3.9	FAR Cleanup & Submittal	2					
4	**Trade Study**	32	16-Feb	23-Feb			
4.1	Trade Study Purpose Definition	3					
4.2	Selection Criteria Development	4					
4.3	Alternatives Analysis	5					
4.4	Criteria Weights Development	4					
4.5	Utility Functions Development	4					
4.6	Raw Criteria Values Collection	4					
4.7	Weighted Utility Calculations	3					
4.8	Sensitivity Analysis	3					
4.9	Trade Study Cleanup & Submittal	2					

WBS ID	Task Name	Planned			Actual		
		Work Hours	Milestone Start	Milestone Finish	Work Hours	Milestone Start	Milestone Finish
5	**Concept Design Report**	37	23-Feb	2-Mar			
5.1	Physical Context Diagram Development	5					
5.2	Physical Component Tree Creation	4					
5.3	Subsystem Descriptions	5					
5.4	Physical Block Diagrams Development	6					
5.5	Physical N2 Diagrams Creation	5					
5.6	Data Flow Diagrams	4					
5.7	Interface Definitions & Descriptions	3					
5.8	Functions-Components Mapping	3					
5.9	Concept Design Cleanup & Submittal	2					
6	**Test Plan**	27	2-Mar	9-Mar			
6.1	Test Objectives Definition	3					
6.2	Requirements-Test Objectives Traceability	4					
6.3	Test Environment Description	3					
6.4	Test Equipment Identification	3					
6.5	Test Subjects & Articles Definition	3					
6.6	Success Criteria Development	3					
6.7	Metrics Definition	3					
6.8	Test Execution Planning	3					
6.9	Test Plan Cleanup & Submittal	2					
7	**A-Specification Report**	22	9-Mar	16-Mar			
7.1	Requirements Update & Organization	4					
7.2	KPP Updates	3					
7.3	Support Requirements Development	3					
7.4	Safety Requirements Development	3					
7.5	HSI Requirements Development	4					
7.6	Requirements-Functions Traceability Update	3					
7.7	A-Spec Cleanup & Submittal	2					
8	**Risk Management Report**	15	16-Mar	30-Mar			
8.1	Risk Definitions Development	2					
8.2	Risk Categorization	2					
8.3	Individual Risk Assessments	3					
8.4	Mitigation Steps Documentation	3					
8.5	Risk Progression Analysis	3					
8.6	Risk Report Cleanup & Submittal	2					
9	**Draft Final Report**	16	30-Mar	6-Apr			
9.1	Executive Summary Development	3					
9.2	Report Sections Integration	4					
9.3	Schedule Assessment & EVM Analysis	3					
9.4	Lessons Learned Documentation	2					
9.5	Next Steps Development	2					
9.6	Draft Final Report Cleanup & Submittal	2					
10	**Final Report & Oral Presentation**	11	6-Apr	25-Apr			
10.1	Final Report Revisions	2					
10.2	Presentation Development	3					
10.3	Presentation Materials Preparation	2					
10.4	Final Report & Presentation Submittal	2					
10.5	Oral Presentation Delivery	2					
		292					

Figure 24: Work Breakdown Structure Update

EVM

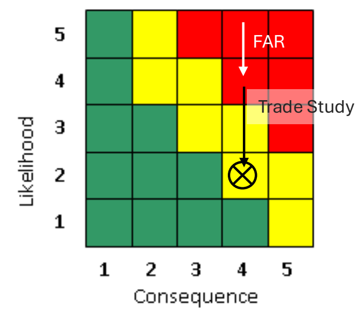
Table 3: EVM Update

					EVM Analysis					
Week	**End of Week Date**	**BCWS**	**BCWP**	**ACWP**	**Cost Variance**	**Schedule Variance**	**SPI**	**CPI**	**TCPI (BAC)**	**EAC**
1	1/26/2025	8	50	5	45	42	6.25	10.00	0.84	29.2
2	2/2/2025	26	50	30	20	24	1.92	1.67	0.92	175.2
3	2/9/2025	44	83.75	55	28.75	39.75	1.90	1.52	0.88	191.8
4	2/16/2025	62	83.75	65	18.75	21.75	1.35	1.29	0.92	226.6
5	2/23/2025	80	83.75	80	3.75	3.75	1.05	1.05	0.98	278.9
6	3/2/2025	98	83.75	100	-16.25	-14.25	0.85	0.84	1.08	348.7
7	3/9/2025	116	83.75	120	-36.25	-32.25	0.72	0.70	1.21	418.4

Appendix B – Risk Updates

Risk 1 – Navigation System Failure

Risk Title	Navigation System Failure	
Description:	IF the GPS integration fails to meet accuracy requirements THEN autonomous positioning during refueling will be compromised, leading to unsafe operating conditions.	
Initial Assessment:	Likelihood:	5
	Consequences:	4
	Description of Consequences if realized	- Autonomous positioning failures would make it difficult to establish rough positioning of the tanker aircraft. Putting the mission, crew members, and others aboard at risk

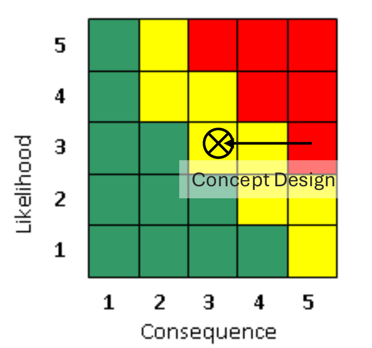


Risk 1: Navigation System Accuracy

Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	5	4	Initial assessment
2	RAR	Impose requirements for implementation of backup navigation system. See: ANGARS-22, Inertial Navigation.	5	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	FAR	Defined F6: Operate in Adverse Conditions behavior	4	4	Likelihood decreases due to implementation of functionality to switch sensors during conditions assessed as adversarial. Further definition of F6.4 should be done to burn down likelihood further.
4	Trade Study	Conducted trade on Navigation System to identify and choose navigation solution which would reduce the likelihood of navigation system failure	2	3	Likelihood reduces significantly. Solution chosen provides mitigation, reliability, redundancy, and accuracy to existing navigation system. Solution does not modify the consequence of risk should the accuracy risk be realized.

Risk 2 – Communications System Failure

Risk Title	Communication System Failure	
Description:	IF secure data link is lost during operation THEN operational delays or mission failure may occur, potentially requiring manual intervention.	
Initial Assessment:	Likelihood:	3
	Consequences:	5
	Description of Consequences if realized	- Comms going down would result in mission failures. Potential critical situational awareness updates may not be conveyed between ANGARS to ground or receiving aircraft, or vise-versa.



Risk 2: Communication System Failure

Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	3	5	Initial assessment
2	RAR	Impose requirements defining design-level communication up-time constraints. See: ANGARS-33, Communication Uptime.	3	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	Concept Design	Imposed requirements to define redundant communication channels/modes for SATCOM and nominal comms. Implemented solution into design space through concept design	2	3	Likelihood that primary comms goes down has not been impacted due to the fact that primary comms is still susceptible to attack from adversaries. Consequence decreases significantly due to the fact that we now have a backup mechanism in the event of an attack on our primary comms mode, we have a secondary antenna which operates via different characteristics.
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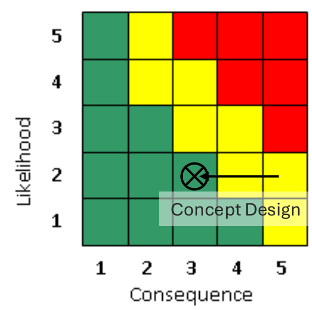
Risk 3 – Collision Avoidance System Failure

Risk Title	Collision Avoidance System Failure	
Description:	IF autonomous collision avoidance fails to activate when needed THEN there is an increased risk of mid-air collisions.	
Initial Assessment:	Likelihood:	2
	Consequences:	5
	Description of Consequences if realized	- Autonomous collision avoidance system failures could result in catastrophic consequences such as bird strikes and fuel boom strikes.



Risk 3: Collision Avoidance Failure

Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	2	5	Initial assessment
2	RAR	Impose requirements defining design-level collision expectations. See: ANGARS-53, Collision Expectancy	2	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	Concept Design	Implementation has been designed and imposed to implement a solution of triple-redundant flight control computers and dual-redundant flight processors	2	3	Consequence reduces significantly due to the fact the implementation of redundant collision avoidance systems and sensors.
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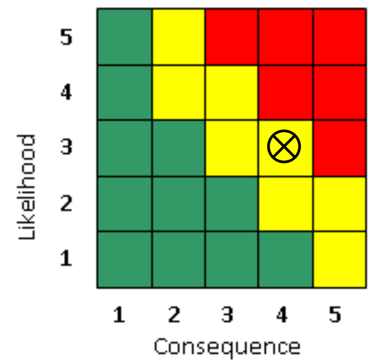
Risk 4 – Integration Challenges

Risk Title	Collision Avoidance System Failure	
Description:	IF there are delays in integration with KC-46/NGAS platforms THEN schedule overruns and increased costs will impact project delivery	
Initial Assessment:	Likelihood:	3
	Consequences:	4
	Description of Consequences if realized	- Integration issues could result in significant schedule and cost risks, including overrun/budget. - Potential implications to existing ANGARS architecture or modifications to legacy hardware.



Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	3	4	Initial assessment
2	RAR	Impose requirements defining design-level integration expectations. See: ANGARS-125, KC-46 Integration	3	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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Appendix C – References

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